

Introduction to Contract Theory and Auctions

Hidden Action: Moral Hazard

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What is Moral Hazard?

- Agent's action (e.g., effort) is not observable (or verifiable by courts).
- But an outcome correlated with effort (e.g., profit) is.
- Also known as **hidden action** model.
- Different from **hidden information** models (screening).
- The term moral hazard comes from an insurance context:
 - ▶ Ability of insured individuals to affect the probabilities of events is called moral hazard.
 - ▶ Insurance industry regarded this as an ethical, or moral, problem.
- Very widespread and recognized phenomena.
 - ▶ Insurance contracts.
 - ▶ Managerial and workers' compensation contracts.
 - ▶ Sharecropping contracts between landlords and tenants.
 - ▶ Financial contracts.
 - ▶ Service industry: car-owner/mechanic or patient/doctor.
 - ▶ Development economics: financial aid to underdeveloped countries.

Moral Hazard

There is moral hazard when

- Agent takes an action that affects his payoff as well as the principal's
 - ▶ How hard to work
- Principal only observes an outcome, an imperfect indicator of the action.
 - ▶ How many items the salesperson sold.
- Agent chooses an inefficient action
 - ▶ Salesperson slacks off.
- Principal's problem is to find a contract that induces high effort.
- If the agent is **risk neutral** (and there is no limited liability) efficient outcome can still be obtained.
- If the agent is **risk averse** then there is a trade off between
 - ▶ **risk sharing**: agent's wage should not depend too strongly on outcome.
 - ▶ **incentives**: to get high effort wage must depend on outcome.

Risk aversion

- Consider the following lotteries over money (no subjective uncertainty, only objective, i.e., single state of the world)
 - ▶ S : I will give you 10,000 Tenge.
 - ▶ R : I will flip a fair coin. If Heads I will give you 20,000 Tenge; if tails I will give you nothing.
- What do you prefer?
 - ▶ If you prefer S over R you are **risk averse**
 - ▶ If you are indifferent you are **risk neutral**
 - ▶ If you prefer R over S you are **risk lover**

An individual is **risk averse** if he prefers to receive the expected value of a lottery to playing the lottery.

Payoff function (of certain outcome) is

- Affine if risk neutral: $u(x) = Ax + B$
- Concave if Risk averse: $u(x) = \sqrt{x}$
- Convex if risk lover: $u(x) = x^2$

A simple example

- The principal will hire a manager for a new project.
- Revenue is
 - ▶ 500K if successful
 - ▶ 0 if not
- Probability of success depends on the manager's effort
 - ▶ High effort: 0.9
 - ▶ Low effort: 0.1
- Effort is costly for the manager
 - ▶ High effort: 250K
 - ▶ Low effort: 50K
- Outside option of the manager is 0

An example: Risk Neutral Manager

- Suppose that the manager is risk neutral.

- If you can observe his effort level

- ▶ pay 250K for high and 0 for low effort.
- ▶ Manager expends high effort
- ▶ Expected profit

$$0.9 \times 500 - 250 = 200$$

- ▶ This is the first best for the principal.

An example: Risk Neutral Manager

- What is the optimal pay schedule if effort cannot be observed?
- Constant wage?
- Manager will choose low effort.
- Then the best wage is 50K
- Expected profit

$$0.1 \times 500 - 50 = 0$$

- Can the principal do better?

An example: Risk Neutral Manager

- 25K plus 250K bonus if successful
- Manager:
 - ▶ Low effort: $25 + 0.1 \times 250 - 50 = 0$
 - ▶ High effort: $25 + 0.9 \times 250 - 250 = 0$
 - ▶ Chooses high effort

- Expected profit

$$0.9 \times (500 - 250) - 25 = 200$$

- Same as observable effort!

An Example: Risk Averse Manager

- If you offer 25K salary plus 250K bonus if successful.
- Expected wage of the manager: 0
- But there is a downside: With positive probability negative payoff.
- Outside option is a sure 0 payoff.
- A risk averse manager does not accept it.
- Principal has to increase expected wage
 - ▶ How much depends on how risk averse the manager is
- To induce high effort there has to be a bonus.
- Profit will be less than the first best.

A Simple Model

- We will consider a simple model with 2 actions and 2 outcomes.
 - ▶ action $a \in \{0, 1\}$ (e.g., low effort, high effort)
 - ▶ outcome $q \in \{0, 1\}$ (failure success)
- Probability of success increases with a
 - ▶ $\Pr(q = 1|a = 0) = p_0 > 0$
 - ▶ $\Pr(q = 1|a = 1) = p_1 > p_0$
- Agent's payoff if he takes action a and gets paid w

$$u(w) - c(a)$$

- ▶ Assume: $u(0) = 0, u' > 0, u'' \leq 0$ and $c(0) = 0 < \psi = c(1)$

- Principal's payoff

$$S(q) - w$$

Let $S(1) = s_1 > s_0 = S(0)$

Complete Information

- Suppose action is observable and verifiable
- Principal can condition payment w on action a
- Agent's participation constraint is the only constraint
 - ▶ Principal decides which effort level is optimal.
 - ▶ If agent does not choose that level he is punished.

- A contract is $w(q) = (w_0, w_1)$

- Participation constraint
 - ▶ If high effort is induced

$$p_1 u(w_1) + (1 - p_1) u(w_0) - \psi \geq 0$$

- ▶ If low effort is induced

$$p_0 u(w_1) + (1 - p_0) u(w_0) \geq 0$$

Complete information

- Suppose the principal wants to induce high effort.
- Her problem can be written as

$$\begin{aligned} \max_{w_0, w_1} & p_1(s_1 - w_1) + (1 - p_1)(s_0 - w_0) \\ \text{subject to} & \\ & p_1 u(w_1) + (1 - p_1)u(w_0) - \psi \geq 0 \end{aligned}$$

- Objective function is linear (hence concave) in w_0 and w_1
- Constraint function is concave as well.
- Kuhn-Tucker conditions are necessary and sufficient for global optimum.

Complete Information

- The Lagrangean is

$$\mathcal{L} = p_1(s_1 - w_1) + (1 - p_1)(s_0 - w_0) + \lambda[p_1 u(w_1) + (1 - p_1)u(w_0) - \psi]$$

- First order conditions

$$w_1: -p_1 + \lambda p_1 u'(w_1) = 0$$

$$w_2: -(1 - p_1) + \lambda(1 - p_1)u'(w_0) = 0$$

$$\lambda: p_1 u(w_1) + (1 - p_1)u(w_0) - \psi \geq 0$$

$$\lambda \geq 0, \quad \lambda[p_1 u(w_1) + (1 - p_1)u(w_0) - \psi] = 0$$

- From the first constraint we must have $\lambda > 0$
 - ▶ Therefore, PC constraint binds.
 - ▶ Makes sense: otherwise we could decrease w

- From the first two constraints

$$\lambda = \frac{1}{u'(w_1)} = \frac{1}{u'(w_0)}$$

which is known as the **Borch rule** after Karl Borch

- Since $u' > 0$, this implies $w_0 = w_1$

Complete Information

- The agent is completely insured.
- A general phenomenon.
 - ▶ Risk neutral party completely insures the risk averse party.
- Exposing agent to risk would require a payment of a risk premium
- If u is strictly concave and $w_0 \neq w_1$

$$u(p_1 w_1 + (1 - p_1) w_0) > p_1 u_1(w_1) + (1 - p_1) u_2(w_0)$$

- Rather than paying agent $w_0 \neq w_1$, pay him a constant wage

$$w = p_1 w_1 + (1 - p_1) w_0 - \epsilon$$

with $\epsilon > 0$ small.

- Can still satisfy PC at a lower expected wage.
- When incentives are not an issue, setting $w_0 \neq w_1$ has no benefit for the principal but it has a cost.

Complete Information

- Optimal wage w can be found using IR constraint.

$$u(w) = \psi$$

- Let $h = u^{-1}$ be the inverse of u
- Note that

$$h'(u) = \frac{1}{u'(h(u))} > 0, \quad h''(u) = -\frac{u''(h(u))h'(u)}{[u'(h(u))]^2} > 0$$

- Then optimal wage is

$$w = h(\psi)$$

- Principal's payoff

$$p_1 s_1 + (1 - p_1) s_0 - h(\psi)$$

Complete Information

- Principal might want to induce low effort $\alpha = 0$
- The problem becomes

$$\max_{w_0, w_1} p_0(s_1 - w_1) + (1 - p_0)(s_0 - w_0)$$

subject to

$$p_0 u(w_1) + (1 - p_0) u(w_0) \geq 0$$

- $u(0) = 0$ and $u' > 0$ imply optimal wage is zero.
- Principal's payoff is $p_0 s_1 + (1 - p_0) s_0$
- Principal prefers to induce high effort if and only if

$$p_1 s_1 + (1 - p_1) s_0 - h(\psi) \geq p_0 s_1 + (1 - p_0) s_0$$

or

$$\underbrace{(p_1 - p_0)(s_1 - s_0)}_{\text{expected benefit}} \geq \underbrace{h(\psi)}_{\text{cost}}$$

Incomplete information

- When the action is not observable or verifiable by a court, the agent needs to be given an incentive to choose high effort.
- Agent's incentive compatibility (IC) constraint must be satisfied

$$p_1 u(w_1) + (1 - p_1) u(w_0) - \psi \geq p_0 u(w_1) + (1 - p_0) u(w_0)$$

- If the principal wants to induce high effort, her problem is

$$\max_{w_0, w_1} p_1 (s_1 - w_1) + (1 - p_1) (s_0 - w_0)$$

$$\text{subject to } p_1 u(w_1) + (1 - p_1) u(w_0) - \psi \geq 0$$

$$p_1 u(w_1) + (1 - p_1) u(w_0) - \psi \geq p_0 u(w_1) + (1 - p_0) u(w_0)$$

- We will analyze this problem for two alternative scenarios
 - ▶ Agent is risk neutral
 - ▶ Agent is risk averse

Incomplete Information: Risk neutral Agent

- Without loss of generality, we can take u to be $u(w) = w$.
- The problem becomes:

$$\begin{aligned} & \max_{w_0, w_1} p_1(s_1 - w_1) + (1 - p_1)(s_0 - w_0) \\ & \text{subject to } p_1 w_1 + (1 - p_1)w_0 - \psi \geq 0 \\ & \quad p_1 w_1 + (1 - p_1)w_0 - \psi \geq p_0 w_1 + (1 - p_0)w_0 \end{aligned}$$

- Can solve using the Kuhn-Tucker conditions (verify this). However, the solution can be found intuitively.
- PC constraint must bind: otherwise we can decrease w_1 and w_0 by the same small amount.
 - ▶ This would increase payoff.
 - ▶ IC compatibility constraint continues to hold

$$(p_1 - p_0)(w_1 - w_0) - \psi \geq 0$$

- Principal's payoff at the solution is:

$$p_1 s_1 + (1 - p_1)s_0 - \psi$$

Incomplete Information: Risk Neutral agent

- Any (w_0, w_1) that satisfies PC with equality and IC is a solution.
- For example, the following at which IC also holds with equality

$$w_0 = -\frac{p_0}{p_1 - p_0}\psi \quad \text{and} \quad w_1 = \frac{1 - p_0}{p_1 - p_0}\psi$$

- Agent is rewarded for success and punished for failure.
- Similarly can solve the problem for low effort.
- If high effort is more desirable under complete information, it is also desirable under incomplete information.
- Moral hazard is not an issue when agent is risk neutral
 - ▶ Principal can induce efficient effort without any extra cost.
 - ▶ There is no extra cost because the agent is risk neutral.
 - ▶ Therefore, giving him the right incentives (by exposing him to risk) is costless.

Incomplete Information: Risk Neutral Agent

- The wage schedule given above is just one possibility
- Can also find other transfer schemes that achieve the same outcome.
- Principal can “sell the store” for

$$T = p_1 s_1 + (1 - p_1) s_0 - \psi$$

- PC constraint holds

$$p_1 s_1 + (1 - p_1) s_0 - \psi - T = 0$$

- IC also holds

$$p_1 s_1 + (1 - p_1) s_0 - \psi - T \geq p_0 s_1 + (1 - p_0) s_0 - T$$

assuming that $(p_1 - p_0)(s_1 - s_0) \geq \psi$.

- In this case, the principal makes the agent a **residual claimant**
 - ▶ And this gives the agent the right incentives.

Incomplete Information: Risk Averse Agent

- If the principal wants to induce high effort, her problem is

$$\begin{aligned} \max_{w_0, w_1} \quad & p_1(s_1 - w_1) + (1 - p_1)(s_0 - w_0) \\ \text{subject to} \quad & p_1 u(w_1) + (1 - p_1)u(w_0) - \psi \geq 0 \\ & (p_1 - p_0)[u(w_1) - u(w_0)] - \psi \geq 0 \end{aligned}$$

- $p_1 > p_0$, $u' > 0$, imply $w_1 > w_0$
- PC constraint must hold as equality: Otherwise can decrease w_0 and w_1 by the same small amount.
 - ▶ PC and IC constraints still hold $w_1 > w_0 \Rightarrow u'(w_1) < u'(w_0)$
 - ▶ Payoff is strictly higher

Incomplete Information: Risk Averse Agent

- The Lagrangean

$$\begin{aligned}\mathcal{L} = & p_1(s_1 - w_1) + (1 - p_1)(s_0 - w_0) + \lambda((p_1 - p_0)[u(w_1) - u(w_0)] - \psi) \\ & + \mu(p_1 u(w_1) + (1 - p_1)u(w_0) - \psi)\end{aligned}$$

- The Kuhn-Tucker Conditions

$$\frac{1}{u'(w_0)} = \mu - \lambda \frac{p_1 - p_0}{1 - p_1} \quad (1)$$

$$\frac{1}{u'(w_1)} = \mu + \lambda \frac{p_1 - p_0}{p_1} \quad (2)$$

$$(p_1 - p_0)[u(w_1) - u(w_0)] - \psi \geq 0 \quad (3)$$

$$p_1 u(w_1) + (1 - p_1)u(w_0) - \psi \geq 0 \quad (4)$$

$$\lambda \geq 0 \quad (5)$$

$$\mu \geq 0 \quad (6)$$

$$\lambda((p_1 - p_0)[u(w_1) - u(w_0)] - \psi) = 0 \quad (7)$$

$$\mu(p_1 u(w_1) + (1 - p_1)u(w_0) - \psi) = 0 \quad (8)$$

Incomplete Information: Risk Averse agent

- Suppose that $\lambda = 0$: (1) and (2) $\Rightarrow w_1 = w_0 \Rightarrow$ contradicts (3).
- Therefore, $\lambda > 0 \Rightarrow$ IC constraint binds.
- $\mu = 0 \Rightarrow$ contradicts (1). Therefore, $\mu > 0 \Rightarrow$ PC constraint binds.
- Can solve for $u(w_0)$ and $u(w_1)$ using IC and PC as equalities.

$$u(w_1) = \psi + \psi \frac{1 - p_1}{p_1 - p_0} = \frac{\psi(1 - p_0)}{p_1 - p_0}$$
$$u(w_0) = \psi - \psi \frac{p_1}{p_1 - p_0} = -\frac{\psi p_0}{p_1 - p_0}$$

Brief Technical point

Recall our discussion about expected utility representations (topic 1). The state independent utility function $u(w)$ represents the same preferences as any **affine** transformation $\hat{u}(w) = Au(w) + B$. This observation will be relevant because the solution for $u(w_0)$ is negative, but we may still want to work with some “usual” utility functions.

Incomplete Information: Risk Averse Agent

- Therefore,

$$w_1 = h\left(\frac{\psi(1-p_0)}{p_1-p_0}\right)$$

$$w_0 = h\left(-\frac{\psi p_0}{p_1-p_0}\right)$$

- Wage if success higher than complete information wage:

$$w_1 > h(\psi)$$

- Wage if failure is lower:

$$w_0 < h(\psi)$$

- Agent is exposed to risk to give him incentives to choose high effort.

Incomplete Information: Risk Averse Agent

- Agent must be paid a risk premium to participate.
- From the PC constraint and strict concavity of u

$$\psi = p_1 u(w_1) + (1 - p_1) u(w_0) < u(p_1 w_1 + (1 - p_1) w_0)$$

- Therefore, second best cost of inducing high effort is higher than the first best cost.

$$p_1 w_1 + (1 - p_1) w_0 > h(\psi)$$

- Incomplete Information makes inducing high effort less desirable.

Incomplete Information: Risk Averse Agent

- How do we know Kuhn-Tucker conditions identify global optimum?
- We will show this by transforming the problem.
 - ▶ This transformation is also useful for practical purposes when solving the problem.
- Define $u_1 = u(w_1)$, $u_0 = u(w_0)$ so that $w_1 = h(u_1)$ and $w_0 = h(u_0)$
- Rewrite the problem as

$$\begin{aligned} \max_{u_1, u_0} \quad & p_1(s_1 - h(u_1)) + (1 - p_1)(s_0 - h(u_0)) \\ \text{subject to} \quad & p_1 u_1 + (1 - p_1) u_0 - \psi \geq 0 \\ & (p_1 - p_0)[u_1 - u_0] - \psi \geq 0 \end{aligned}$$

- Now objective function is strictly concave
 - ▶ That is because h is strictly convex
- Constraints are linear, thus they identify a convex feasible set (actually a convex polytope).
- **Technical point:** A constraint qualification holds (for example, Slater's condition, $\exists(u_0, u_1) \in \mathbb{R}_+^2$ s.t. constraints hold as strict inequalities)
- Kuhn-Tucker conditions identify (the unique) global maximum.

Limited Liability

- In some situations there are constraints on the transfers between principal and agent.
 - ▶ Wages cannot be negative.
 - ▶ Borrower can declare bankruptcy
- These situations are known as **limited liability**.
- Suppose that transfers to agent must be greater than $-l$, where $l \geq 0$.
- For simplicity assume agent is risk neutral.

Limited Liability

- The problem becomes

$$\begin{aligned} \max_{w_0, w_1} \quad & p_1(s_1 - w_1) + (1 - p_1)(s_0 - w_0) \\ \text{subject to} \quad & p_1 w_1 + (1 - p_1)w_0 - \psi \geq 0 \\ & (p_1 - p_0)[w_1 - w_0] - \psi \geq 0 \\ & w_0 \geq -l \\ & w_1 \geq -l \end{aligned}$$

- IC implies $w_1 > w_0$. therefore, $w_0 \geq -l$ implies $w_1 > -l$
 - ▶ We can ignore $w_1 \geq -l$ constraint.
- Either PC or $w_0 \geq -l$ must bind. Otherwise
 - ▶ Decrease w_0 and w_1 by the same small amount and increase payoff.

Limited Liability

- Suppose: $l < p_0\psi/(p_1 - p_0)$

- ▶ $w_0 = -l \Rightarrow$ IC binds.

$$w_0 = -l, \quad w_1 = -l + \frac{\psi}{p_1 - p_0}$$

- Suppose: $l \geq p_0\psi/(p_1 - p_0)$

- ▶ $w_0 > -l \Rightarrow$ PC binds.

- ▶ One solution is where IC binds too

$$w_0 = -\frac{p_0}{p_1 - p_0}\psi \quad \text{and} \quad w_1 = \frac{1 - p_0}{p_1 - p_0}\psi$$

Limited Liability

- Therefore, if $l < \psi \frac{p_0}{p_1 - p_0}$ the solution is given by

$$w_0 = -l, \quad w_1 = -l + \frac{\psi}{p_1 - p_0}$$

- Compare with no limited liability solution

$$w_0 = -\frac{p_0}{p_1 - p_0}\psi \quad \text{and} \quad w_1 = \frac{1 - p_0}{p_1 - p_0}\psi$$

- Principal cannot punish agent sufficiently in case of failure.
- To create incentives the principal has to reward more in case of success.
- This leaves the agent some rent and reduces principal's payoff.
- Limited liability makes inducing high effort less desirable.

More than Two Outcomes

- There are n levels of performance $q_1 < q_2 < \dots < q_n$.
- Still two action levels: 0 and 1.
- $\Pr(q = q_i | a = 0) = p_{i0} > 0$
- $\Pr(q = q_i | a = 1) = p_{i1} > 0$
 - ▶ The distribution of performance under $a = 1$ **first-order stochastically dominates** the distribution under $a = 0$, i.e., for every $j \in \{1, \dots, n\}$ we have $\sum_{i=1}^j p_{i0} \geq \sum_{i=1}^j p_{i1}$
- Principal's payoff $S(q_i) - w$, $s_i = S(q_i)$, $j > i \Rightarrow s_j > s_i$
- A contract is $(w_1, w_2, \dots, w_n)$
- PC constraint for high effort

$$\sum_{i=1}^n p_{i1} u(w_i) - \psi \geq 0$$

- IC constraint for high effort

$$\sum_{i=1}^n p_{i1} u(w_i) - \psi \geq \sum_{i=1}^n p_{i0} u(w_i)$$

- The analysis will be similar to the two outcome case.

Complete Information

- Only PC constraint needs to be satisfied, which binds at the solution.
- Efficient risk sharing

$$\frac{1}{u'(w_i)} = \frac{1}{u'(w_j)}$$

- Constant wage w such that

$$\sum_{i=1}^n p_{i1} u(w) = \psi$$

or

$$w = h(\psi)$$

as in the case of two outcomes.

Incomplete Information

- FOC is similar to two outcome case: for each $i = 1, \dots, n$

$$\frac{1}{u'(w_i)} = \mu + \lambda \left(\frac{p_{i1} - p_{i0}}{p_{i1}} \right)$$

where μ is the Lagrange multiplier for PC and λ is the multiplier for IC constraint.

- Multiplying both sides of FOC by p_{i1} and summing over i yields

$$\mu = \sum_{i=1}^n p_{i1} \frac{1}{u'(w_i)} = E \left[\frac{1}{u'(w)} \right] > 0$$

where E is the expectation operator.

- This implies that PC binds.
- Can also show $\lambda > 0$, i.e., IC binds.
 - ▶ $\lambda = 0 \Rightarrow w_i = w_j, \forall i, j$, which violates IC.

Incomplete Information

- In two outcome case we had $w_1 > w_0$
- This is intuitive: to induce high effort you pay more in case of higher output, which is more likely under high effort.
- Is this still true? Is it the case that $j \geq i \Rightarrow w_j \geq w_i$?
- Since $u'' < 0$, using the FOC we can show that this is the case if and only if

$$\frac{p_{j1}}{p_{j0}} \geq \frac{p_{i1}}{p_{i0}}$$

- These ratios are called **Likelihood ratios**
- Therefore, monotonicity of wage schedule holds under the assumption of **monotone likelihood property** (MLRP)

$$\frac{p_{i1}}{p_{i0}} \text{ is increasing in } i$$

Monotone Likelihood Ratio

- The principal's problem is partly statistical.
 - ▶ Given that you observe q_j , action 1 is likelier to have occurred than action 0 if $p_{j1} > p_{j0}$
 - ▶ A measure of how much “likelier” is the likelihood ratio.

$$\frac{p_{j1}}{p_{j0}}$$

- ▶ The higher this ratio the stronger an evidence j is for action i
 - ▶ MLRP says that higher output levels are stronger evidence for high effort ($a = 1$).
- If MLRP holds, rewarding higher output levels with higher payments is the optimal way of providing incentives to work harder.

General Model

- When there are more than two effort levels the problem becomes quite difficult to solve.
- General results are difficult to obtain.
- For example, MLRP is not sufficient for monotonicity.
 - ▶ It is sufficient together with a condition known as **convexity of distribution function**
- For an analysis of a general model see Grossman, S. and Hart, O. (1983)